COMET: COUNTERFACTUAL MOMENT AUDITING UNDER WEAK OVERLAP AND DEPENDENT FEEDBACK

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ABSTRACT

Logged decision systems increasingly combine learned policies, adaptive data collection, and interacting units. Their counterfactual statistics can be unstable for three distinct reasons: target and logging policies may have only weak overlap, outcomes may be heavy-tailed, and observations may be dependent through repeated users, slates, networks, or temporal feedback. We introduce COMET, a general framework for estimating and testing families of counterfactual moments in this regime. The framework begins with an orthogonal score that covers off-policy value, subgroup effects, counterfactual calibration, safety constraints, and exposure-mediated spillovers. It then combines dependence-separated cross-fitting with graph–time robust aggregation to produce simultaneous certificates over target policies and auditors. The proposed rate separates heavy-tail difficulty, an observable conflict-graph design effect, nuisance-estimation error, temporal coupling, and within-window drift. Building on these certificates, we formulate a discover–certify procedure for finding hidden failure subgroups without reusing the certification data, and a pessimistic policy-repair rule with high-probability feasibility. We specify a complete empirical program spanning controlled synthetic processes, semi-synthetic temporal graphs, and real logged recommendation data. The resulting problem is broader than counterfactual fairness on evolving graphs: that setting is recovered as one instance of a common counterfactual-moment interface.

1 INTRODUCTION

Modern decision systems learn from logs generated by earlier versions of themselves. Recommendation, advertising, marketplace ranking, experimentation, and clinical decision support all use a logging policy to select an exposure or action, observe only the corresponding outcome, and then update a model or policy. The scientific questions asked of those logs are broader than average policy value. Practitioners also need to know whether a score remains calibrated under an intervention, whether a target policy violates a safety or resource constraint, whether treatment effects differ across hidden subpopulations, and whether exposure produces unequal direct or spillover effects. Each question can be written as a *counterfactual moment*: an expectation of an intervention-specific orthogonal score multiplied by an auditor or constraint function.

Reliable estimation of these moments is difficult when three pathologies interact. First, overlap may be weak. A target policy can place mass on actions that the logging policy chose rarely, producing large importance ratios even when outcomes are bounded. Strong overlap assumptions hide this difficulty and can fail precisely for the policies that are most useful to evaluate. Second, the outcome itself may be heavy-tailed: watch time, transaction value, medical cost, and user-level aggregate engagement are not naturally bounded. Third, logged units are rarely independent. Candidates share a slate, repeated observations share users, neighboring nodes interfere, and policy updates induce temporal dependence. Treating such data as independent can yield narrow intervals with severe undercoverage.

Existing solutions address important pieces of this problem. Doubly robust estimators reduce sensitivity to nuisance estimation (Dudík et al., 2011; Jiang and Li, 2016; Farajtabar et al., 2018); weak-overlap analyses characterize when truncation is necessary (Wang et al., 2017; Mehrabi and Wager, 2024; Khan et al., 2024); robust mean estimators tolerate finite moments (Lugosi and Mendelson, 2019); and causal inference under interference formalizes exposure mappings and dependency structures (Aronow and Samii, 2017; Sävje et al., 2021). What is missing is a common

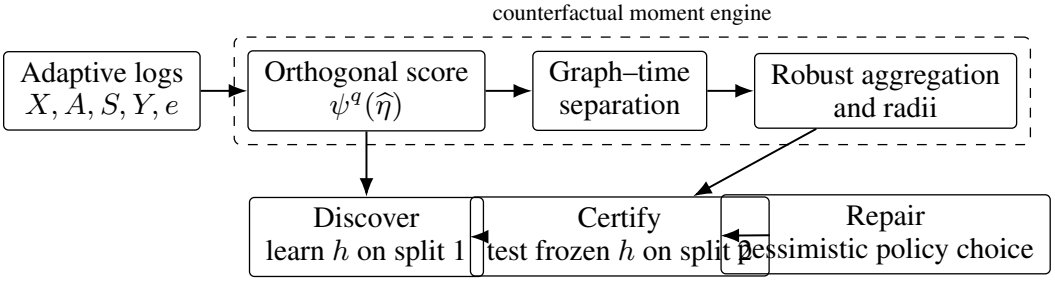


Figure 1: COMETseparates the counterfactual score, the dependence-aware robust estimator, and downstream use. Rich auditors may be learned on a discovery split, but certification and policy repair use a separate graph–time split.

interface that simultaneously supports a *family* of counterfactual questions, learned nuisance functions, heavy-tailed scores, and observable graph–time dependence—and then turns the resulting uncertainty into decisions rather than a post-hoc error bar.

We introduce **COMET** (**CO**unterfactual **M**oment **E**stimation and **T**esting). COMEThas a modular structure. A generalized exposure mapping converts a possibly joint action into a local exposure state. A cross-fitted orthogonal score converts the partially observed outcome into a pseudo-outcome for a target intervention. A graph–time separation step prevents nuisance leakage and records which scores can be treated as independent or weakly coupled. Finally, a heavy-tail robust aggregation module produces simultaneous confidence radii over a policy family and an auditor family. The same engine can estimate policy value, certify counterfactual calibration, compare subgroup effects, test spillovers, or enforce a cost constraint.

The statistical engine supports a broader workflow, illustrated in Figure 1. On a discovery split, a flexible auditor searches for a subgroup with a large counterfactual residual. The selected auditor is frozen and evaluated on a dependence-separated certification split, avoiding the double use of data that otherwise turns subgroup search into an invalid significance test. The resulting upper and lower confidence bounds can then be used in a pessimistic policy-selection problem: utility is optimized through a lower confidence bound while every risk or fairness constraint is enforced through an upper confidence bound. We refer to this workflow as *discover–certify–repair*.

Our formulation is deliberately not tied to a single fairness metric or graph task. Counterfactual online performative fairness on evolving graphs (Li and Zou, 2026) is recovered by choosing a shown/not-shown exposure mapping, group–score auditors, and exposure-effect constraints. In the present work, this is one application rather than the organizing premise. The generalization changes both the assumptions and the scientific question: strict overlap is replaced by finite-moment weak overlap; fixed hand-designed slices are augmented by learned auditors; a stationary population mean is replaced by a window target with explicit drift; and a generic stochastic optimizer is replaced by certificate-guided policy repair.

Contributions. The paper develops the following components.

1. **A counterfactual-moment interface.** We give one orthogonal-score formulation that covers off-policy value, subgroup value, counterfactual calibration, treatment-effect heterogeneity, safety and resource constraints, and local spillovers.
2. **Dependence-aware robust certification.** We combine graph–time separated cross-fitting with robust block aggregation. The resulting target bound decomposes into a finite-p heavy-tail term with rate $N_{\mathrm{eff}}^{-(p-1)/p}$, a nuisance-product remainder, a temporal coupling term, and a within-window drift term. The dependence penalty is based on a computable conflict graph rather than an unobserved statement that samples are "approximately i.i.d."
3. **Discover–then–certify auditing.** A rich model can search for a violating subgroup on one split; a frozen auditor is then certified on another. Ratio metrics include uncertainty in both the numerator and subgroup mass and abstain when support is insufficient.

4. **Certificate-guided policy repair.** We formulate pessimistic constrained policy selection and show that simultaneous confidence bounds imply high-probability feasibility for the selected policy.
5. **A falsifiable empirical program.** Experiments are designed around coverage, interval width, false certification, detection power, effective-sample-size scaling, subgroup recovery, and utility–constraint tradeoffs on synthetic, semi-synthetic, and real logged data. No conclusion relies only on point-estimation error.

2 Counterfactual Moments from Adaptive Dependent Logs

2.1 Logged process and local exposure

At round $t = 1, \dots, T$, a system observes the history $\mathcal{F}_{t-1}$ and a set of decision units $\mathcal{I}_t$. A unit $u = (t, i)$ has context X_u, while the system draws a possibly joint action vector A_t from a predictable logging policy. A known *exposure mapping*

$$S_u = g_u\big(A_{I(u)}\big) \in \mathcal{S} \tag{1}$$

compresses the relevant part of the joint assignment into a finite local exposure state. Examples include a binary shown/not-shown indicator, an item's position in a slate, a treatment together with the fraction of treated neighbors, or a discretized dose. For every $s \in \mathcal{S}$, let $Y_u(s)$ be the potential outcome under local exposure s; the observed outcome is $Y_u = Y_u(S_u)$. We allow Y_u to be unbounded.

The generalized propensity and outcome regression are

$$e_u(s) := \mathbb{P}(S_u = s \mid X_u, \mathcal{F}_{t-1}), \qquad \mu_u(s) := \mathbb{E}[Y_u(s) \mid X_u, \mathcal{F}_{t-1}]. \tag{2}$$

A target intervention $q \in \mathcal{Q}$ is a predictable conditional distribution $q_u(s \mid X_u, \mathcal{F}_{t-1})$. It may represent a new ranking policy, a stochastic treatment rule, or a local intervention used only for auditing.

Assumption 2.1 (Local identification)**.** For every evaluated unit and exposure state: (i) consistency holds through Equation (1); (ii) $Y_u(s) \perp S_u \mid X_u, \mathcal{F}_{t-1}$; and (iii) $q_u(s) > 0$ implies $e_u(s) > 0$ almost surely.

The third condition requires support only where the target intervention places mass. It does *not* impose a constant lower bound on $e_u(s)$.

2.2 Orthogonal scores and moment families

For nuisance functions $\eta = (\mu, e)$, define the generalized doubly robust score

$$\psi_u^q(\eta) := \sum_{s \in \mathcal{S}} q_u(s) \left[\mu_u(s) + \frac{\mathbf{1}\{S_u = s\}}{e_u(s)} \{Y_u - \mu_u(s)\} \right]. \tag{3}$$

Let $h \in \mathcal{H}$ be an auditor or constraint function, and let f_u be a predictable reference quantity such as a model score, a budget level, or zero. For a window $\mathcal{I}_W = \cup_{t \in W} \mathcal{I}_t$ of size N_W, our target is the pathwise average predictable moment

$$M_W(q, h; f) := \frac{1}{N_W} \sum_{u \in \mathcal{I}_W} \mathbb{E}\big[h_u\{\psi_u^q(\eta_0) - f_u\} \mid \mathcal{F}_{t(u)-1}\big], \tag{4}$$

where $h_u = h(X_u, \mathcal{F}_{t(u)-1})$ and $\eta_0 = (\mu_0, e_0)$. The target is allowed to depend on the realized history. It therefore describes the counterfactual behavior of the system along the deployment path actually being audited, rather than assuming that the adaptive process has reached a stationary distribution.

Proposition 2.2 (Identification and double robustness)**.** *Under Theorem 2.1,* $\mathbb{E}[\psi_u^q(\eta_0) \mid X_u, \mathcal{F}_{t-1}] = \sum_s q_u(s)\mu_u(s)$. *Moreover, for generic nuisances* $\bar{\eta} = (\bar{\mu}, \bar{e})$,

$$\mathbb{E}[\psi_u^q(\bar{\eta}) - \psi_u^q(\eta_0) \mid X_u, \mathcal{F}_{t-1}] = \sum_s q_u(s)\{\bar{\mu}_u(s) - \mu_u(s)\} \frac{\bar{e}_u(s) - e_u(s)}{\bar{e}_u(s)}. \tag{5}$$

Thus the conditional bias vanishes if either nuisance component is correct and otherwise is second order in their errors.

Task	Choice of target/reference	Auditor or interpretation
Policy value	q target policy, $f = 0$	$h \equiv 1$
Subgroup value	q target policy, $f = 0$	$h = \mathbf{1}\{G = g\}$ or a learned subgroup
Counterfactual calibration	q intervention, $f = \widehat{p}(X)$	score-bin/group indicators or signed calibration tests
Effect heterogeneity	contrast $\psi^{q_1} - \psi^{q_0}$	tests of conditional treatment or exposure effects
Safety/resource constraint	outcome is cost, harm, or load	$M_W(q, h) \le \alpha_h$
Local spillover	$S = (A_i, \bar{A}_{N(i)})$	direct and neighbor-exposure contrasts
COPF instance	shown/not-shown S	group–score auditors and exposure-effect parity

Table 1: Examples covered by the counterfactual-moment interface. A single run may certify many target policies and auditors simultaneously.

The same moment interface yields several common tasks.

2.3 Weak overlap and heavy-tailed scores

Large scores can arise from small propensities, heavy-tailed outcomes, or both. We encode the statistically relevant object directly rather than assuming bounded outcomes and bounded importance ratios.

Assumption 2.3 (Finite-p counterfactual moments)**.** For some $p \in (1, 2]$ and $\nu_p < \infty$, uniformly over $(q, h) \in \mathcal{Q} \times \mathcal{H}$,

$$\frac{1}{N_W} \sum_{u \in \mathcal{I}_W} \mathbb{E}\big[|h_u\{\psi_u^q(\eta_0) - f_u\}|^p \mid \mathcal{F}_{t(u)-1}\big] \le \nu_p^p. \tag{6}$$

No second moment is required when $p < 2$.

A sufficient condition controls the weighted residual term $\sum_s q_u(s)^p e_u(s)^{1-p} \mathbb{E}[|Y_u(s) - \mu_u(s)|^p \mid X_u, \mathcal{F}_{t-1}]$. This makes the weak-overlap mechanism explicit: even bounded outcomes can generate finite-p but infinite-variance scores when $e_u(s)$ approaches zero sufficiently often. Conversely, strong overlap does not eliminate heavy tails when the outcome residual is heavy-tailed. Experiments in Section 6 vary these two mechanisms separately.

2.4 Observable dependence and drift

Within a window, define a conflict graph $G_W = (\mathcal{I}_W, E_W)$. Two vertices are adjacent whenever their scores may share randomized exposure, outcome noise, a repeated entity, or an interference neighborhood. The graph is constructed from logged identifiers, slate membership, time, and the specified exposure mapping. Let χ_W denote the number of colors returned by a fixed proper-coloring algorithm and d_W its degeneracy. Both are observable design quantities; $\chi_W \le d_W + 1 \le \Delta(G_W) + 1$.

Temporal dependence remains possible across conflict-free units. We therefore partition time into macro-blocks separated by a gap ℓ. For an offline β-mixing analysis, a Berbee-style coupling contributes a failure-probability term of order $B\beta(\ell)$ for B retained blocks. In adaptive deployments, we instead condition on the past and treat the block targets as predictable. To make nonstationarity explicit, let

$$\Delta_W(q, h) := \max_{b \le B} |M_b(q, h) - M_W(q, h)| \tag{7}$$

measure within-window target drift. Shorter windows reduce Δ_W but also reduce effective sample size, producing an auditable bias–variance–drift tradeoff.

3 The COMET Estimator

COMETseparates three operations that are often conflated: nuisance learning, dependence control, and heavy-tail robustification. This separation is useful both statistically and operationally: logging

propensities can be audited independently of the outcome model, the conflict graph can be inspected before seeing outcomes, and the robust aggregation rule can be stress-tested without retraining the backbone policy.

3.1 Dependence-separated cross-fitting

Random row-wise cross-fitting is invalid when rows share a user, slate, neighborhood, or nearby time state. We construct evaluation folds at the macro-block level and add a temporal guard band. For fold k, nuisances $\widehat{\eta}^{(-k)} = (\widehat{\mu}^{(-k)}, \widehat{e}^{(-k)})$ are trained only on blocks that are outside the guard band and have no prohibited entity overlap with the evaluation fold. If the platform logs exact propensities, $\widehat{e} = e$ and only the outcome regression is learned. The cross-fitted score for u in fold $k(u)$ is

$$\widehat{Z}_u(q,h) := h_u \left\{ \psi_u^q\left(\widehat{\eta}^{(-k(u))}\right) - f_u \right\}. \tag{8}$$

Fold construction uses only design variables available before the outcome is observed. This prevents outcome-adaptive removal of difficult observations.

3.2 Graph–time robust aggregation

Let $b = 1, \dots, B$ index temporally separated evaluation blocks. For the clean estimator and theorem, blocks are constructed with a common retained size n_0 (the final incomplete block is dropped), so $N_W = Bn_0$. A deterministic proper coloring gives χ_b color classes and we set

$$\bar{\chi}_W := \max_{b \le B} \chi_b, \qquad N_{\text{eff}} := \frac{N_W}{\bar{\chi}_W}. \tag{9}$$

This is a conservative effective sample size. A clique block contributes roughly one independent observation per n_0 rows, while an edgeless block contributes all n_0 rows. Degeneracy $d_b + 1$ can replace χ_b when a fast greedy coloring is used. A variable-block weighted extension is given in the appendix, but the equal-block construction avoids changing the target population through precision weighting.

For each (q, h), compute the block mean

$$\bar{Z}_b(q,h) := \frac{1}{n_b} \sum_{u \in b} \widehat{Z}_u(q,h). \tag{10}$$

Blocks are partitioned into K super-blocks with the same number of temporal blocks (up to one). The gth mean is

$$\widetilde{Z}_g(q,h) := \frac{1}{|g|} \sum_{b \in g} \bar{Z}_b(q,h), \tag{11}$$

and the primary estimate is

$$\widehat{M}_W(q,h) := \text{median}_{g \le K}\, \widetilde{Z}_g(q,h). \tag{12}$$

This is a graph-aware median-of-means (MOM) construction. The block representation allows heavy-tail robustness to operate at the same granularity used to control temporal dependence. An optional streaming variant replaces the median by adaptive score truncation; it is described in Section H.

Tail tuning. The cleanest theorem treats p and an upper scale ν_p as known. The implementation will expose three modes: (i) an oracle mode used only to verify rate predictions in synthetic data; (ii) a sensitivity grid over p and robust scales, reporting the union of valid intervals; and (iii) a data-adaptive Lepski-style mode that selects among nested truncation/MOM scales. The adaptive mode is a required empirical component and a candidate secondary theorem, not an assumption hidden in the implementation.

3.3 Simultaneous radii and diagnostic decomposition

For a finite family $\Theta = \mathcal{Q} \times \mathcal{H}$, the working radius has the form

$$\begin{aligned} r_W(q,h;\delta) = C_p \widehat{\nu}_p(q,h) \left(\frac{\log(2\,|\Theta|\,/\delta)}{N_{\text{eff}}} \right)^{\frac{p-1}{p}} + \widehat{R}_{\text{nuis}}(q,h) \\ + \widehat{R}_{\text{coup}}(q,h) + \widehat{R}_{\text{drift}}(q,h). \end{aligned} \tag{13}$$

Algorithm 1 COMET certification for a finite policy–auditor family

Require: Logged data; target policies $\mathcal{Q}$; auditors $\mathcal{H}$; exposure map g; tail index p or a tuning grid; confidence δ.

1: Build time macro-blocks, guard bands, and conflict graphs from design variables.
2: Construct dependence-separated cross-fitting folds; fit $\widehat{\eta}^{(-k)}$.
3: **for** each evaluation unit u and $(q, h) \in \mathcal{Q} \times \mathcal{H}$ **do**
4: Compute $\widehat{Z}_u(q, h)$ from Equation (8).
5: **end for**
6: Retain equal-size blocks; color each conflict graph; compute $\bar{\chi}_W$ and $N_{\text{eff}} = N_W/\bar{\chi}_W$.
7: Form equal-count super-blocks and the robust estimates in Equation (12).
8: Compute simultaneous radii $r_W(q, h; \delta)$, including nuisance, coupling, and drift allowances.
9: **return** $[\widehat{M}_W - r_W, \widehat{M}_W + r_W]$ for every (q, h), plus diagnostics $(N_{\text{eff}}, \chi_b,$ overlap tails, abstentions$)$.

The four terms answer different debugging questions. A large heavy-tail term indicates insufficient effective support; a large nuisance term points to model or propensity error; a large coupling term calls for wider temporal spacing; and a large drift term calls for shorter or explicitly nonstationary windows. Reporting only the total radius would obscure these distinct failure modes.

The exact nuisance bias suggested by Equation (5) is

$$R_{\text{nuis}}(q, h) := \frac{1}{N_W} \sum_u \mathbb{E}\left[|h_u| \sum_s q_u(s) \, |\widehat{\mu}_u(s) - \mu_u(s)| \, \frac{|\widehat{e}_u(s) - e_u(s)|}{\widehat{e}_u(s)} \right]. \tag{14}$$

In experiments it is estimated by cross-fit surrogates and stress-tested under controlled nuisance misspecification. When propensities are logged exactly, this term is zero regardless of outcome-regression error.

3.4 RATIO CERTIFICATES AND ABSTENTION

Many subgroup metrics are ratios. Let $m_h = \mathbb{E}[h_u r_u]$ be a counterfactual residual numerator and $\pi_h = \mathbb{E}[h_u]$ the slice mass. Suppose simultaneous intervals satisfy

$$|m_h - \widehat{m}_h| \le r_{m,h}, \qquad |\pi_h - \widehat{\pi}_h| \le r_{\pi,h}. \tag{15}$$

Then, whenever $\widehat{\pi}_h - r_{\pi,h} > 0$,

$$\frac{|m_h|}{\pi_h} \le \frac{|\widehat{m}_h| + r_{m,h}}{\widehat{\pi}_h - r_{\pi,h}}. \tag{16}$$

If the denominator lower bound is nonpositive, COMET returns *insufficient support to certify*. This abstention is essential under weak overlap and prevents a tiny subgroup from receiving a numerically precise but statistically meaningless fairness or calibration score.

3.5 COMPUTATIONAL INTERFACE

For fixed score arrays, all moments are matrix products between an auditor matrix and counterfactual pseudo-outcomes. Conflict-graph construction is performed once per window; greedy degeneracy coloring is linear in the graph size. The expensive component is typically nuisance fitting, not robust aggregation. The implementation will cache cross-fitted scores so that new hand-designed auditors can be certified without retraining, while learned auditors must respect the discovery/certification split in Section 5.

4 FINITE-SAMPLE GUARANTEES

We state the core result for a finite family $\Theta = \mathcal{Q} \times \mathcal{H}$. Extensions to VC-type auditor classes can replace $\log |\Theta|$ by an entropy integral, but the finite-family statement is already sufficient for policy libraries, score buckets, and the certification stage of a learned auditor.

4.1 A GRAPH MOMENT REDUCTION

The following elementary reduction explains why the coloring penalty enters with the same exponent as sample size.

Lemma 4.1 (Finite-p moment under a conflict graph)**.** *Let $X_1, \dots, X_n$ be conditionally mean-zero random variables with a conditional dependency graph that admits a proper coloring with χ colors. If $p \in (1, 2]$ and $\mathbb{E}[|X_i|^p \mid \mathcal{D}] \le \nu^p$, then*

$$\mathbb{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n} X_i\right|^p \middle| \mathcal{D}\right] \le C_p \nu^p \left(\frac{\chi}{n}\right)^{p-1}, \tag{17}$$

where $\mathcal{D}$ contains the design variables and coloring.

The proof partitions the variables by color, applies the von Bahr–Esseen inequality within each independent color class, and combines the color sums using convexity. Thus n/χ behaves as a conservative effective sample size without requiring every graph neighborhood to be uniformly small.

4.2 UNIFORM ROBUST CERTIFICATE

Let $\bar{Z}_b(\theta)$ be the oracle-score mean in temporal block b for $\theta = (q, h)$. The retained blocks have common size n_0 and satisfy $\chi_b \le \bar{\chi}_W$. Assume blocks can be coupled to independent copies with total coupling failure at most $\varepsilon_{\text{coup}}$. This includes geometrically β-mixing blocks separated by a gap ℓ, for which $\varepsilon_{\text{coup}}$ can be bounded by a sum of mixing coefficients. Split the B blocks into K MOM super-blocks of equal cardinality up to one.

Theorem 4.2 (Uniform COMET certificate)**.** *Suppose Theorems 2.1 and 2.3 hold, the nuisance functions are fitted on dependence-separated folds, the centered cross-fitted scores obey the same conditional p-moment envelope ν_p, and Theorem 4.1 applies conditionally within every evaluation block. Set*

$$K = \left\lceil 8 \log \frac{2\,|\Theta|}{\delta} \right\rceil, \qquad K \le B/2. \tag{18}$$

Then there is a constant C_p depending only on p such that, with probability at least $1 - \delta - \varepsilon_{\text{coup}}$,

$$\sup_{\theta=(q,h)\in\Theta} \left|\widehat{M}_W(\theta) - M_W(\theta)\right| \le C_p \nu_p \left(\frac{\log(2\,|\Theta|\,/\delta)}{N_{\text{eff}}}\right)^{\frac{p-1}{p}} + R^\star_{\text{nuis}} + \Delta^\star_W, \tag{19}$$

where $R^\star_{\text{nuis}} = \sup_\theta R_{\text{nuis}}(\theta)$ and $\Delta^\star_W = \sup_\theta \Delta_W(\theta)$. If block coupling is exact and the target is stable within the window, the last probability and error corrections vanish.

At $p = 2$, the stochastic term is $N_{\text{eff}}^{-1/2}$ up to logarithms. For $p < 2$, it becomes $N_{\text{eff}}^{-(p-1)/p}$, which is the canonical finite-p mean-estimation rate. The theorem makes three distinctions that are important in practice. First, N_{eff} is based on the logged conflict graph rather than the raw number of candidate rows. Second, weak overlap and heavy-tailed outcomes enter through the same score moment ν_p but can be diagnosed separately from the propensity and residual distributions. Third, adaptivity is not hidden inside a stationary expectation: nonstationarity appears as the measurable window-drift term $\Delta^\star_W$.

Proof idea. Condition on the cross-fitting training data and the design. Theorem 4.1 converts each block mean into an observation whose pth centered moment is at most a constant times $\nu_p^p(\bar{\chi}_W/n_0)^{p-1}$. Averaging B/K independent temporal blocks gives a super-block moment of order $\nu_p^p(K/N_{\text{eff}})^{p-1}$, so Markov's inequality makes each super-block accurate with a constant probability. A binomial tail bound implies that a majority of super-blocks are accurate, and a union bound over Θ yields the logarithmic complexity. Equation (5) contributes the nuisance-product remainder. The coupling event transfers the result from independent block copies to the original sequence, while nonidentical block targets contribute at most $\Delta^\star_W$. A detailed derivation is provided in Section D.

Corollary 4.3 (Certified ratio moments)**.** *On the event in Theorem 4.2, any ratio metric $|m_h|/\pi_h$ obeys Equation (16) whenever the lower confidence bound on π_h is positive. Otherwise, abstention is the only conclusion guaranteed by the data.*

4.3 A MATCHING DEPENDENCE–TAIL LOWER BOUND

The coloring penalty is not merely an artifact of the proof. Consider a subclass in which N observations form N/χ independent clusters of size χ, and every observation within a cluster is an exact copy. This is a valid dependency graph with clique number and chromatic number χ.

Theorem 4.4 (Cluster-replication lower bound)**.** *Fix $p \in (1, 2]$ and suppose χ divides N. Over distributions satisfying $\mathbb{E}\,|X - \mathbb{E}X|^p \le \nu_p^p$ and the cluster-replication dependence model above, every estimator $\widetilde{M}$ satisfies*

$$\inf_{\widetilde{M}} \sup_{P} \mathbb{E}_P \left| \widetilde{M} - M(P) \right| \ge c_p \nu_p \left(\frac{\chi}{N} \right)^{\frac{p-1}{p}} . \tag{20}$$

The proof reduces the problem to mean estimation from N/χ independent finite-p samples and applies a two-point heavy-tail construction. Consequently, the interaction between local dependence and tail index in Equation (19) is minimax unavoidable for the general class, up to logarithmic factors. Sharper graph parameters may improve over a particular coloring for structured graphs, but no method can replace effective sample size by raw sample size without additional assumptions.

4.4 SCOPE OF THE GUARANTEE

The theorem certifies an identified counterfactual moment; it cannot diagnose violations of ignorability or an incorrect exposure mapping from logged outcomes alone. It also separates, rather than eliminates, temporal coupling and drift. A strong final version should sharpen the observable design effect using fractional colorings or graph-specific variance proxies and prove validity of the adaptive tail-tuning rule. These refinements do not change the problem definition or the downstream discover–certify–repair interface.

5 DISCOVER, CERTIFY, AND REPAIR

5.1 LEARNING FAILURE SUBGROUPS WITHOUT INVALIDATING INFERENCE

Hand-designed group and score buckets are interpretable but can miss interactions among user history, item type, time, model confidence, and network position. Directly maximizing an empirical violation over a rich auditor class and reporting the same value as a certificate is invalid: the search overfits the noise, especially when inverse propensities are heavy-tailed.

COMET uses a two-stage protocol. Split the window into graph–time separated discovery and certification sets. On the discovery set, fit an auditor

$$\widehat{h} \in \arg\max_{h \in \mathcal{H}_{\mathrm{rich}}} \left| \widehat{M}_{\mathrm{disc}}(q, h) \right| - \lambda\, \Omega(h), \tag{21}$$

where $\mathcal{H}_{\mathrm{rich}}$ may be a shallow tree, sparse rule list, boosted model, or neural auditor and Ω controls complexity or minimum support. Freeze $\widehat{h}$, then run Algorithm 1 on the certification split. Conditional on the discovery data, $\widehat{h}$ is a fixed test, so the certification guarantee does not depend on the complexity of the search model. Multiple discovered auditors can be certified by including the frozen top-L candidates in a finite family and paying only $\log L$ in the radius.

The protocol separates two questions that are often confused: *Can a model find a suspicious pattern?* and *Does fresh dependent data support that pattern counterfactually?* A useful learned auditor should succeed on both.

5.2 PESSIMISTIC POLICY REPAIR

Let $V(q)$ be a counterfactual utility moment and $C_j(q)$, $j = 1, \ldots, J$, be risk, cost, calibration, spillover, or fairness moments with acceptable levels α_j. Suppose COMET gives simultaneous estimates and radii over a candidate library $\mathcal{Q}$:

$$\left| \widehat{V}(q) - V(q) \right| \le r_V(q), \qquad \left| \widehat{C}_j(q) - C_j(q) \right| \le r_j(q). \tag{22}$$

Algorithm 2 Discover–certify auditing

Require: A target intervention q, a rich auditor class, and graph–time separated discovery/certification data.

1: Fit or cache cross-fitted counterfactual scores on the discovery split.
2: Learn one or more supported auditors using Equation (21); freeze their parameters and textual definitions.
3: Recompute nuisance functions and scores using only training data permitted for the certification split.
4: Certify each frozen auditor with COMET, including numerator and support-mass uncertainty.
5: **return** certified violations, non-violations, and abstentions; retain the discovery score only as a search diagnostic.

We select

$$\widehat{q} \in \arg\max_{q \in \mathcal{Q}} \{\widehat{V}(q) - r_V(q)\} \quad \text{s.t.} \quad \widehat{C}_j(q) + r_j(q) \leq \alpha_j \; \forall j. \tag{23}$$

This rule is pessimistic in both directions: it optimizes a utility lower confidence bound and constrains risk upper confidence bounds. The policy library may contain the current policy, mixtures with an exploration policy, threshold or group-offset variants, and policies produced by an unconstrained learner. A continuous neural policy class can be optimized first, but only a frozen candidate library is used in the certification theorem.

Proposition 5.1 (Feasibility and utility of repaired policy)**.** *On the simultaneous event Equation (22), any solution of Equation (23) satisfies $C_j(\widehat{q}) \leq \alpha_j$ for all j. Moreover, for every comparator q satisfying the robust margin condition $C_j(q) + 2r_j(q) \leq \alpha_j$ for all j,*

$$V(\widehat{q}) \geq V(q) - 2r_V(q). \tag{24}$$

The margin condition is informative rather than cosmetic. A policy whose estimated constraint lies exactly at the boundary cannot be certified without additional data. As N_{eff} grows, the robustly feasible set expands and the utility penalty contracts at the heavy-tail rate. This connects statistical design choices—exploration, candidate budget, interference radius, window length—directly to deployable policy quality.

Sequential deployment. In an online system, repair is run at auditable epochs rather than after every event. A new policy is admitted only if its current-window certificates pass; exploration and propensity logging remain active after deployment. Sequential validity can be obtained by allocating confidence across epochs or replacing fixed-window bounds with e-processes, but full anytime-valid theory is left as an extension. The experiments report cumulative violations and detection delay under controlled shifts rather than claiming fixed-window intervals are automatically time-uniform.

6 EXPERIMENTAL DESIGN

The empirical goal is not to demonstrate that one estimator has the smallest point error under a single simulator. The central claim is about *reliable certification*; therefore coverage and false certification are primary endpoints. We organize the study around five questions.

RQ1 Do nominal 90% and 95% intervals attain empirical coverage under weak overlap, heavy-tailed outcomes, and dependence?

RQ2 Does error scale with $N_{\text{eff}}^{-(p-1)/p}$ rather than raw row count, and is the dependence penalty necessary?

RQ3 Which component matters in which regime: orthogonality, heavy-tail robustification, graph correction, temporal separation, or drift control?

RQ4 Can discover–certify recover an unknown violating subgroup while controlling false discoveries?

RQ5 Does pessimistic repair reduce true constraint violations with a competitive utility–safety frontier?

6.1 CONTROLLED DYNAMIC PROCESS

We use two graph families: a dynamic degree-corrected stochastic block model for controlled community structure and a preferential-attachment stream for hub-heavy dependence. At round t, a user/source is sampled, a candidate set is constructed, and the behavior policy selects an item or Top-K slate according to

$$\pi_t = (1 - \epsilon_{\text{exp}})\,\text{Softmax}(s_t/\tau) + \epsilon_{\text{exp}}\,\text{UniformTopK}\,. \tag{25}$$

A sticky latent regime Z_t induces temporal persistence. For unit $u = (t, i)$, outcomes follow

$$Y_u = \mu(X_u, Z_t) + \tau_{G_u} A_u + \rho_{G_u} \overline{A}_{N(u),t} + b_{\text{entity}(u)} + \varepsilon_u, \tag{26}$$

where τ_g is a direct exposure effect, ρ_g is a spillover effect, and b_{entity} creates repeated-user dependence. The noise is Gaussian, Student-t, or Pareto/Lomax with a controlled tail index. We evaluate four regimes: light-tailed control; weak overlap only; heavy-tailed outcome only; and both mechanisms together. Ground-truth moments are computed analytically when possible and otherwise by a large independent Monte Carlo intervention run whose error is at least an order of magnitude below the plotted intervals.

Sweeps vary sample size, tail index p, exploration ϵ_{exp}, candidate budget, Top-K, hub strength, interference radius, repeated-entity frequency, temporal persistence, window length, auditor count, and nuisance misspecification. The key rate plot places median absolute error and interval width against both N and N_{eff} on log–log axes. The predicted slope is $-(p-1)/p$ against N_{eff}; fitting a slope against raw N is a negative control.

6.2 REAL STRUCTURES AND LOGGED INTERVENTIONS

The datasets play complementary roles rather than being treated as interchangeable benchmarks.

TGB semi-synthetic. Temporal Graph Benchmark (TGB) provides large real graph streams and standardized temporal splits (Huang et al., 2023). We preserve timestamps, repeated nodes, degree heterogeneity, and candidate construction, then simulate the logging policy and potential-outcome mechanism on top of the observed stream. This yields exact counterfactual truth while retaining realistic conflict graphs. The main tests compare no dependence correction, temporal-only blocks, graph-only correction, and the full graph–time design. We also compare maximum-degree, degeneracy, and greedy-coloring proxies for effective sample size.

Open Bandit Dataset. Open Bandit Dataset contains logged recommendations collected by multiple behavior policies on the same e-commerce platform, including action probabilities and outcomes (Saito et al., 2021). One policy's log is used to estimate the value of another, while the latter's on-policy sample average provides an external evaluation target. Because the reward is primarily binary, this experiment isolates heavy tails induced by importance ratios rather than by the outcome. Target policies are also interpolated between the logged policies to create a graded overlap path.

KuaiRand. KuaiRand provides sequential recommendation logs with user identifiers, timestamps, multiple feedback fields, and randomly exposed videos inserted into standard feeds (Gao et al., 2022). It is used to study repeated-user dependence, drift, and heavy-tailed watch-time outcomes. The randomized-exposure portion supports held-out intervention checks, but we do not claim exact ground truth for every arbitrary target policy. Evaluation therefore emphasizes stability across random-exposure validation slices, support-aware abstention, and subgroup replication over time.

6.3 BASELINES AND ABLATIONS

Estimator baselines include direct method (DM), IPS, self-normalized IPS, ordinary DR, clipped DR, switch/truncated DR, more-robust DR, i.i.d. MOM-DR, temporal block-MOM, graph-corrected non-robust DR, and COMET. For learned subgroup search we compare fixed slices, unconstrained in-sample search, sample-split search without dependence separation, and discover–certify. Policy repair baselines are utility-only selection, empirical constrained selection without radii, clipped-DR constraints, and the COPF decision-layer controller in the evolving-link application (Li and Zou, 2026).

Setting	Exact truth	Weak overlap	Outcome tail	Graph/time dep.	Primary purpose
Synthetic DCSBM/PA	Yes	Controlled	Controlled	Controlled	Coverage, rates, lower-bound regimes
TGB semi-synthetic	Yes	Controlled	Controlled	Real structure	Dependence correction and N_{eff}
Open Bandit	On-policy target	Real/graded	Limited	Mild temporal	Real propensity OPE
KuaiRand	Partial validation	Real	Watch time	Repeated users/time	Drift, abstention, subgroup replication

Table 2: The evaluation suite separates exact counterfactual validation from realism. No single dataset is asked to establish every claim.

Method	Coverage ↑	Width ↓	False cert. ↓	Repair violation ↓
Ordinary DR	–	–	–	–
Clipped / truncated DR	–	–	–	–
i.i.d. MOM-DR	–	–	–	–
Temporal block-MOM	–	–	–	–
Graph-aware non-robust	–	–	–	–
COMET	–	–	–	–

Table 3: Main-result table shell. Values are intentionally blank until the preregistered experiment scripts are run. The final paper should report uncertainty over repetitions, not only means.

Ablations remove one component at a time: outcome regression, exact/logged propensities, graph edges, temporal gaps, robust aggregation, nuisance separation, denominator uncertainty, and drift allowance. To expose silent failure, every method must return both an estimate and either an interval or an explicit statement that it has no valid interval under the tested assumptions.

6.4 METRICS AND REPORTING

For estimation we report bias, RMSE, median absolute error, empirical coverage, average interval width, and worst-case coverage across policies/auditors. For certification we report false certification rate, power, abstention rate, and support-stratified performance. For subgroup discovery we report overlap with the planted subgroup, effect-size regret relative to the best supported subgroup, and replication on the certification split. For repair we report true utility, maximum true constraint violation, cumulative violation over deployment epochs, and detection delay after a shift. Runtime, peak memory, graph-construction time, and effective-sample-size diagnostics are reported separately.

All synthetic experiments use at least enough independent repetitions to place a two-sided binomial confidence interval of width at most two percentage points around nominal 95% coverage. Hyperparameters are selected without using counterfactual ground truth, except in explicitly labeled oracle-tail experiments. Dataset preprocessing, candidate construction, policy mixtures, auditor definitions, and random seeds are fixed in configuration files before the final comparison.

7 RELATED WORK

Off-policy and orthogonal estimation. Inverse-propensity and doubly robust estimators are standard tools for contextual bandits and sequential decision problems (Dudík et al., 2011; Jiang and Li, 2016; Wang et al., 2017; Farajtabar et al., 2018). Cross-fitting and Neyman orthogonality allow flexible nuisance learners while keeping first-order target inference insensitive to small nuisance errors (Chernozhukov et al., 2018; Foster and Syrgkanis, 2019). COMET adopts this orthogonal-score principle but changes the inferential object from one average policy value to a simultaneous family

of policy–auditor moments, and changes the sampling model from independent bounded scores to weak-overlap, heavy-tailed, graph–time dependent logs.

Weak overlap and heavy tails. Strong overlap bounds the importance ratio uniformly, while weak-overlap work studies slower rates, truncation, or partial identification when this condition fails (Mehrabi and Wager, 2024; Khan et al., 2024). Robust mean estimation provides exponential-confidence procedures under finite moments through median-of-means, truncation, and related estimators (Lugosi and Mendelson, 2019). The contribution here is not the use of MOM in isolation. It is the combination of orthogonal counterfactual scores, an observable dependence design effect, simultaneous auditing, nuisance remainders, and downstream certified policy selection. Our experiments separately vary tail behavior from the outcome and from the importance ratio because the same empirical score tail can have different operational remedies.

Interference and dependent data. Exposure mappings and design-based estimators formalize causal effects when one unit's outcome depends on others' assignments (Aronow and Samii, 2017; Sävje et al., 2021). Dependency-graph concentration controls sums of partly dependent variables by graph coloring or related fractional parameters (Janson, 2004); blocking and coupling are classical tools for mixing sequences (Merlevède et al., 2009). COMET uses these ideas as a logged-system interface: conflict edges are generated by slates, repeated entities, and declared interference neighborhoods, and the resulting design penalty is reported with every certificate. This differs from assuming an abstract effective sample size without showing how it is computed.

Auditing, calibration, and performative systems. Multicalibration asks predictions to be calibrated over many subpopulations (Hébert-Johnson et al., 2018), while outcome indistinguishability gives a broader test-based view of prediction quality (Dwork et al., 2021). Performative prediction studies how deployment changes the data distribution (Perdomo et al., 2020). COPF applies counterfactual exposure effects, residual OI, and online control to evolving-graph link recommendation (Li and Zou, 2026). COMET retains the useful residual-moment abstraction but removes the task-specific dependence on link prediction, fixed fairness slices, strict overlap, and one controller. Learned auditors are separated from certification, and policy repair is formulated directly through simultaneous uncertainty bounds.

Benchmarks. Open Bandit Dataset enables reproducible OPE comparisons using real logged propensities and multiple behavior policies (Saito et al., 2021). KuaiRand adds randomized exposures and rich sequential recommendation feedback (Gao et al., 2022). TGB supplies realistic temporal graph structure and standardized streams (Huang et al., 2023). We use them for distinct validation roles, supplemented by controlled generators in which the full counterfactual truth and dependence graph are known.

8 CONCLUSION

This paper reframes robust counterfactual fairness on evolving graphs as a more general problem: estimating, testing, and acting on counterfactual moments from weak-overlap, heavy-tailed, dependent logs. COMET supplies one orthogonal-score interface, an observable graph–time correction, robust simultaneous certificates, a sample-split mechanism for discovering hidden failure subgroups, and a pessimistic policy-repair rule. The framework makes uncertainty operational: inadequate overlap, dependence, nuisance error, and drift widen different parts of the certificate or force abstention. The remaining work is concrete and falsifiable—complete the sharp dependence proof and adaptive tail tuning, run coverage-centered experiments, and determine whether the predicted effective-sample-size scaling holds on realistic graph and recommendation logs.

REPRODUCIBILITY STATEMENT

The estimands, assumptions, algorithms, and evaluation metrics are stated in the main paper. The appendix provides derivations, proof blueprints, dependence-construction details, synthetic data-generating processes, dataset-specific protocols, and a checklist of claims that must be verified before submission. The final version will include an anonymous code package containing data preprocessing,

simulation, estimator implementations, and scripts that reproduce every table and figure from fixed configuration files and random seeds.

Ethics Statement

The framework is intended to diagnose and constrain counterfactual failures in deployed decision systems, but a valid certificate is conditional on the exposure mapping, logged propensities, identification assumptions, auditor family, and data quality. Misspecified interference neighborhoods or unmeasured confounding can create false assurance. Sensitive group labels should be collected and processed only with appropriate legal, organizational, and privacy safeguards. Dataset experiments will follow the licenses and intended-use conditions of the original releases. Policy repair should not be treated as a substitute for domain review, stakeholder consultation, or prospective evaluation.

AI Use Statement

Generative AI tools were used in research ideation and to help reorganize prior technical notes, draft portions of the prose and LaTeX, propose mathematical formulations, and outline theoretical and experimental directions. The authors are responsible for independently checking every definition, proof, citation, implementation, and empirical claim; unverified suggestions must not be presented as established results.

References

Peter M. Aronow and Cyrus Samii. Estimating average causal effects under general interference, with application to a social network experiment. *The Annals of Applied Statistics*, 11(4):1912–1947, 2017. doi: 10.1214/16-AOAS1005.

Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018. doi: 10.1111/ectj.12097.

Miroslav Dudík, John Langford, and Lihong Li. Doubly robust policy evaluation and learning. In *Proceedings of the 28th International Conference on Machine Learning*, pages 1097–1104, 2011.

Cynthia Dwork, Michael P. Kim, Omer Reingold, Guy N. Rothblum, and Gal Yona. Outcome indistinguishability. In *Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing*, pages 1095–1108. ACM, 2021. doi: 10.1145/3406325.3451064.

Mehrdad Farajtabar, Yinlam Chow, and Mohammad Ghavamzadeh. More robust doubly robust off-policy evaluation. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1447–1456. PMLR, 2018.

Dylan J. Foster and Vasilis Syrgkanis. Orthogonal statistical learning. In *Proceedings of the 32nd Conference on Learning Theory*, volume 99 of *Proceedings of Machine Learning Research*, pages 1362–1385. PMLR, 2019.

Chongming Gao, Shijun Li, Yuan Zhang, Jiawei Chen, Biao Li, Wenqiang Lei, Peng Jiang, and Xiangnan He. Kuairand: An unbiased sequential recommendation dataset with randomly exposed videos. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, pages 3953–3957. ACM, 2022. doi: 10.1145/3511808.3557624.

Úrsula Hébert-Johnson, Michael P. Kim, Omer Reingold, and Guy N. Rothblum. Multicalibration: Calibration for the (computationally-identifiable) masses. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1944–1953. PMLR, 2018.

Shenyang Huang, Farimah Poursafaei, Jacob Danovitch, Matthias Fey, Weihua Hu, Emanuele Rossi, Jure Leskovec, Michael Bronstein, Guillaume Rabusseau, and Reihaneh Rabbany. Temporal graph benchmark for machine learning on temporal graphs. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Svante Janson. Large deviations for sums of partly dependent random variables. *Random Structures & Algorithms*, 24(3):234–248, 2004. doi: 10.1002/rsa.20008.

Nan Jiang and Lihong Li. Doubly robust off-policy value evaluation for reinforcement learning. In *Proceedings of the 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 652–661. PMLR, 2016.

Samir Khan, Martin Saveski, and Johan Ugander. Off-policy evaluation beyond overlap: Sharp partial identification under smoothness. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 23734–23757. PMLR, 2024.

Sheng'en Li and Dongmian Zou. Counterfactual online performative fairness. *Proceedings of the 43rd International Conference on Machine Learning*, 2026. Accepted at ICML 2026; arXiv:2606.00700.

Gábor Lugosi and Shahar Mendelson. Mean estimation and regression under heavy-tailed distributions: A survey. *Foundations of Computational Mathematics*, 19(5):1145–1190, 2019. doi: 10.1007/s10208-019-09427-x.

Mohammad Mehrabi and Stefan Wager. Off-policy evaluation in markov decision processes under weak distributional overlap. *arXiv preprint arXiv:2402.08201*, 2024.

Florence Merlevède, Magda Peligrad, and Emmanuel Rio. Bernstein inequality and moderate deviations under strong mixing conditions. In *High Dimensional Probability V: The Luminy Volume*, volume 5 of *Institute of Mathematical Statistics Collections*, pages 273–292. Institute of Mathematical Statistics, 2009. doi: 10.1214/09-IMSCOLL518.

Juan Perdomo, Tijana Zrnic, Celestine Mendler-Dünner, and Moritz Hardt. Performative prediction. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7599–7609. PMLR, 2020.

Yuta Saito, Shunsuke Aihara, Megumi Matsutani, and Yusuke Narita. Open bandit dataset and pipeline: Towards realistic and reproducible off-policy evaluation. In *Advances in Neural Information Processing Systems: Datasets and Benchmarks Track*, volume 34, 2021.

Fredrik Sävje, Peter M. Aronow, and Michael G. Hudgens. Average treatment effects in the presence of unknown interference. *The Annals of Statistics*, 49(2):673–701, 2021. doi: 10.1214/20-AOS1973.

Yu-Xiang Wang, Alekh Agarwal, and Miroslav Dudík. Optimal and adaptive off-policy evaluation in contextual bandits. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 3589–3597. PMLR, 2017.

A Extended Notation and Application Instantiations

This appendix expands the compact interface in Table 1. For binary action $A \in \{0, 1\}$ without interference, set $S = A$. For a target treatment rule $q(1 \mid X)$, Equation (3) is the familiar augmented inverse-propensity score. For a slate, S may be a position, a shown/not-shown indicator, or a coarser exposure summary. For network interference, $S = (A_i, b(\{A_j : j \in N(i)\}))$, where b discretizes neighbor exposure. The validity of every estimand is relative to this mapping: two assignments treated as the same exposure state are assumed to induce the same potential outcome for the unit.

Off-policy value. Set $h \equiv 1$ and $f = 0$. Then $M_W(q, 1; 0)$ is the window-average value of target intervention q along the observed context path.

Counterfactual calibration. Let $f_u = \widehat{p}_u$ be a deployed score and let q specify the intervention under which calibration is desired. For a group/score slice $h_{g,I} = \mathbf{1}\{G = g, \widehat{p} \in I\}$, the ratio $\mathbb{E}[h_{g,I}(\psi^q - \widehat{p})]/\mathbb{E}[h_{g,I}]$ is a signed counterfactual calibration residual. Absolute-value and maximum-over-slice metrics follow from simultaneous numerator and denominator bounds.

Heterogeneous direct or spillover effects. For two interventions q_1, q_0, replace ψ^q by $\psi^{q_1} - \psi^{q_0}$. Auditors can be protected-group indicators, network roles, or learned subgroup functions. Direct and spillover contrasts are obtained by changing only one coordinate of a multicomponent exposure state.

Safety and resource use. Let Y be cost, harm, latency, or resource consumption and set $f = \alpha$. A one-sided moment certificate determines whether the target policy satisfies a constraint. The same policy can be evaluated against many costs and subgroups without redefining the estimator.

B Identification and Orthogonality

Proof of Theorem 2.2. Condition on $(X_u, \mathcal{F}_{t-1})$ and suppress the unit index. Under consistency and local ignorability,

$$\mathbb{E}\left[\frac{\mathbf{1}\{S=s\}}{e(s)}\{Y-\mu(s)\}\middle| X, \mathcal{F}_{t-1}\right] = \frac{\mathbb{P}(S=s \mid X, \mathcal{F}_{t-1})}{e(s)}\mathbb{E}[Y(s)-\mu(s) \mid X, \mathcal{F}_{t-1}]$$
$$= 0.$$

Summing with weights $q(s)$ yields $\mathbb{E}[\psi^q(\eta_0) \mid X, \mathcal{F}_{t-1}] = \sum_s q(s)\mu(s)$. For generic $\bar{\eta}$, the conditional expectation of the augmentation term is

$$\mathbb{E}\left[\frac{\mathbf{1}\{S=s\}}{\bar{e}(s)}\{Y-\bar{\mu}(s)\}\middle| X, \mathcal{F}_{t-1}\right] = \frac{e(s)}{\bar{e}(s)}\{\mu(s)-\bar{\mu}(s)\}.$$

Adding the regression term and subtracting the oracle target gives

$$\bar{\mu}(s)-\mu(s)+\frac{e(s)}{\bar{e}(s)}\{\mu(s)-\bar{\mu}(s)\} = \{\bar{\mu}(s)-\mu(s)\}\frac{\bar{e}(s)-e(s)}{\bar{e}(s)}.$$

Multiplying by $q(s)$ and summing proves Equation (5). □

A useful deterministic consequence is

$$|\mathbb{E}[h\{\psi^q(\bar{\eta})-\psi^q(\eta_0)\}]| \le \mathbb{E}\left[|h|\sum_s q(s)\,|\bar{\mu}(s)-\mu(s)|\,\frac{|\bar{e}(s)-e(s)|}{\bar{e}(s)}\right], \tag{27}$$

which motivates Equation (14). Cross-fitting makes $\bar{\eta}$ fixed relative to the evaluation fold when applying concentration. It does not by itself guarantee that this remainder is small; that requires nuisance quality in norms weighted by the target policy and inverse propensity.

C Proof of the Conflict-Graph Moment Lemma

Proof of Theorem 4.1. Condition on $\mathcal{D}$ and let $C_1, \dots, C_\chi$ be the color classes. Write $S_c = \sum_{i \in C_c} X_i$. Variables inside a color class are conditionally independent and mean zero. For $1 < p \le 2$, the von Bahr–Esseen inequality gives

$$\mathbb{E}[|S_c|^p \mid \mathcal{D}] \le C_p \sum_{i \in C_c} \mathbb{E}[|X_i|^p \mid \mathcal{D}] \le C_p\,|C_c|\,\nu^p. \tag{28}$$

The color sums need not be mutually independent. Convexity nevertheless implies

$$\left|\sum_{c=1}^{\chi} S_c\right|^p \le \chi^{p-1}\sum_{c=1}^{\chi}|S_c|^p\,. \tag{29}$$

Taking conditional expectations, summing $|C_c| = n$, and dividing by n^p yields

$$\mathbb{E}\left[\left|n^{-1}\sum_i X_i\right|^p \mid \mathcal{D}\right] \le C_p\nu^p\chi^{p-1}n^{1-p}, \tag{30}$$

which is Equation (17). □

The bound is conservative because it uses an integral coloring and ignores cancellations across colors. A fractional coloring or a graph-specific moment inequality may replace χ by a smaller parameter; this is a theorem-strengthening opportunity rather than a change to the estimator interface.

D Proof Blueprint for the Uniform Comet Bound

We provide the complete argument under the equal-block coupled model used in Theorem 4.2. The remaining technical work for the final paper is to state the weakest conditional coupling assumptions and track constants for the chosen adaptive scale estimator.

Fix $\theta \in \Theta$ and condition on the design and nuisance-training data. Let $\mu_b(\theta) = \mathbb{E}[\bar{Z}_b(\theta)]$ for a block of size n_0. By Theorem 4.1,

$$\mathbb{E}\left|\bar{Z}_b(\theta) - \mu_b(\theta)\right|^p \le C_p \nu_p^p \left(\frac{\bar{\chi}_W}{n_0}\right)^{p-1}. \tag{31}$$

On the coupling event, the B block means can be replaced by independent copies with the same marginals. Partition them into K super-blocks of size $L = \lfloor B/K \rfloor$; at most $K-1$ excess blocks can be discarded or deterministically distributed without changing constants. For group g, define

$$\widetilde{Z}_g = \frac{1}{L}\sum_{b\in g} \bar{Z}_b, \qquad \mu_g = \frac{1}{L}\sum_{b\in g} \mu_b. \tag{32}$$

The von Bahr–Esseen inequality across independent temporal blocks and Equation (31) imply

$$\mathbb{E}\left|\widetilde{Z}_g - \mu_g\right|^p \le \frac{C_p}{L^p}\sum_{b\in g}\mathbb{E}\left|\bar{Z}_b - \mu_b\right|^p \tag{33}$$

$$\le C_p' \nu_p^p \left(\frac{\bar{\chi}_W}{L n_0}\right)^{p-1} \le C_p'' \nu_p^p \left(\frac{K}{N_{\text{eff}}}\right)^{p-1}. \tag{34}$$

Choose

$$a = C_p''' \nu_p \left(\frac{K}{N_{\text{eff}}}\right)^{(p-1)/p} \tag{35}$$

with a constant making $\mathbb{P}(\left|\widetilde{Z}_g - \mu_g\right| > a) \le 1/8$ by Markov's inequality. Since super-blocks are independent on the coupling event, a Chernoff bound gives

$$\mathbb{P}\left(\text{at least } K/2 \text{ groups are bad}\right) \le e^{-cK}. \tag{36}$$

The window target is the average of equal-size block targets. By Equation (7), $|\mu_g - M_W| \le \Delta_W$. Hence, whenever a majority of groups are good, their estimates all lie in $[M_W - a - \Delta_W, M_W + a + \Delta_W]$, and so does their median. Taking $K \ge 8\log(2\left|\Theta\right|/\delta)$ and union bounding over Θ proves the stochastic and drift terms.

The argument above is centered at the expectation of the cross-fitted plug-in score. Proposition 2.2 bounds its difference from the oracle counterfactual target by R_{nuis}. Finally, the independent-block result transfers to the original blocks outside a coupling event of probability $\varepsilon_{\text{coup}}$. Combining the events gives Equation (19).

Variable block sizes. A valid extension can preserve the raw-row target by constructing super-blocks with equal total raw mass and using raw-mass weights inside each group. Its moment bound depends on

$$\max_g \frac{\sum_{b\in g} n_b^p (\chi_b/n_b)^{p-1}}{(\sum_{b\in g} n_b)^p} = \max_g \frac{\sum_{b\in g} n_b \chi_b^{p-1}}{(\sum_{b\in g} n_b)^p}. \tag{37}$$

The equal-block estimator in the main text is preferred for the first implementation because its target and design effect are immediately auditable.

E Nuisance Learning under Dependence

A cross-fitting fold is defined by temporal blocks, entity clusters, and conflict edges rather than rows. The following rules prevent common leakage modes.

1. All observations from an entity that can carry latent state across time belong to the same side of a fold boundary or are separated by a declared washout period.

2. Units whose exposure neighborhoods overlap are not split between nuisance training and evaluation when that overlap would reveal the evaluation assignment or outcome.
3. Features are timestamped and constructed only from information available before the logged action.
4. Propensities are taken from the logging mechanism when available; an estimated propensity is evaluated on held-out folds and its lower-tail calibration is reported.

For theoretical statements, the cross-fitted score must have a finite conditional p-moment. Exact logged propensities make this condition easier to diagnose. Estimated propensities can introduce artificial near-zero denominators even when the true behavior policy has adequate support. The implementation therefore reports quantiles of both logged and estimated target-to-logging ratios and treats numerical zeros as support failures. Propensity clipping is included as a baseline, not silently built into the estimand. Robustifying the final orthogonal score is distinct from asserting that the propensity ratio is bounded.

A Cauchy–Schwarz version of the nuisance remainder is available when suitable weighted second moments exist. Define

$$A_q^2 = \mathbb{E}\left[|h| \sum_s q(s)\{\widehat{\mu}(s) - \mu(s)\}^2 w_s\right],$$
$$B_q^2 = \mathbb{E}\left[|h| \sum_s q(s) \frac{\{\widehat{e}(s) - e(s)\}^2}{\widehat{e}(s)^2 w_s}\right]$$

for positive weights w_s. Then $R_{\mathrm{nuis}}(q, h) \le A_q B_q$. Under weak overlap, choosing and estimating these weighted norms is more informative than reporting unweighted nuisance RMSE.

F Temporal Coupling and Conflict-Graph Construction

F.1 Conflict edges

An edge is added between two scored units when any of the following holds under the declared data-generating interface: they share a randomized slate; one lies in the other's interference neighborhood; they share a user or item state not absorbed by the context; their outcomes share a cluster shock; or their feature construction uses a common post-action random variable. Edges are not inferred from outcome correlation after the fact. This keeps the design penalty checkable and prevents outcome-dependent graph selection.

F.2 A protocol-level degree bound

The original evolving-graph application motivates a useful design corollary. Suppose each round processes at most M scored candidate units and an evaluated unit can interact only with units from L_{int} previous or future rounds. If the within-round protocol creates at most cM conflicts for each unit, then

$$\Delta(G_W) \le c(2L_{\mathrm{int}} + 1)M, \qquad \bar{\chi}_W \le c(2L_{\mathrm{int}} + 1)M + 1. \tag{38}$$

This is a protocol bound, not a claim that the underlying graph has bounded degree. Hubs can exist; capping candidate processing and interference memory still controls the logged instance graph. The experiment varying M exposes the resulting compute–statistics tradeoff.

F.3 Mixing blocks

For a stationary β-mixing sequence of block summaries, blocks separated by ℓ can be coupled to independent copies with total failure probability bounded by a constant times $(B-1)\beta(\ell)$. Geometric decay $\beta(\ell) \le C\rho^{\ell}$ yields a logarithmic guard length for a specified coupling budget. In an adaptive system, stationarity may be implausible; the main text therefore reports window drift separately and treats the mixing result as an offline corollary rather than the definition of the target.

G Heavy-Tail Lower Bound

Proof of Theorem 4.4. Let $n = N/\chi$ be the number of independent clusters. Every cluster contains χ identical copies of a scalar X_j, so observing all N rows is statistically equivalent to observing $X_1, \dots, X_n$.

Consider P_0 concentrated at zero and P_1 assigning probability $\varepsilon = 1/n$ to $a = \nu_p \varepsilon^{-1/p}$ and probability $1 - \varepsilon$ to zero. Their means differ by

$$\Delta = \varepsilon a = \nu_p n^{-(p-1)/p}. \tag{39}$$

Moreover, P_1 satisfies $\mathbb{E}_{P_1} |X - \mathbb{E}X|^p \le \mathbb{E}_{P_1} |X|^p = \varepsilon a^p = \nu_p^p$. The total variation distance between P_0^n and P_1^n is

$$1 - (1 - 1/n)^n \le 1 - e^{-1}/2 \tag{40}$$

for all sufficiently large n; finite small n are absorbed into the constant. Le Cam's two-point method therefore gives minimax absolute-error risk at least a constant multiple of Δ. Substituting $n = N/\chi$ proves Equation (20). □

H Streaming Truncation Variant

For online monitoring, storing enough blocks for a MOM estimate may be undesirable. Define a score-truncated estimator

$$\widehat{M}_\tau(q, h) = \frac{1}{N_W} \sum_{u \in \mathcal{I}_W} \widehat{Z}_u(q, h) \mathbf{1}\{\left|\widehat{Z}_u(q, h)\right| \le \tau\}. \tag{41}$$

Under a conditional p-moment bound, truncation bias is at most $\nu_p^p \tau^{-(p-1)}$. Applying a bounded dependency-graph/mixing inequality to the truncated score gives a deviation term that balances at

$$\tau \asymp \left(\frac{\nu_p^p N_{\text{eff}}}{\log(|\Theta| / \delta)} \right)^{1/p}, \tag{42}$$

leading to the same $\nu_p \{\log(|\Theta| / \delta)/N_{\text{eff}}\}^{(p-1)/p}$ rate up to coupling and nuisance corrections. This variant estimates a path-average mean even when block targets vary, but its scale selection is more sensitive and its finite-sample constants can be worse than MOM. It is therefore useful as a streaming method and an ablation, not as an excuse to claim that the method uses no truncation anywhere.

I Policy-Repair Guarantee

Proof of Theorem 5.1. On the simultaneous event, every selected constraint satisfies

$$C_j(\widehat{q}) \le \widehat{C}_j(\widehat{q}) + r_j(\widehat{q}) \le \alpha_j. \tag{43}$$

Let q satisfy $C_j(q) + 2r_j(q) \le \alpha_j$. Then

$$\widehat{C}_j(q) + r_j(q) \le C_j(q) + 2r_j(q) \le \alpha_j, \tag{44}$$

so q is feasible for Equation (23). Optimality of $\widehat{q}$ and the utility confidence bounds imply

$$\begin{aligned} V(\widehat{q}) &\ge \widehat{V}(\widehat{q}) - r_V(\widehat{q}) \\ &\ge \widehat{V}(q) - r_V(q) \\ &\ge V(q) - 2r_V(q). \end{aligned}$$

□

J Detailed Experimental Protocol

J.1 Synthetic parameters

The initial grid should include $p \in \{1.2, 1.5, 1.8, 2.0\}$; exploration $\epsilon_{\text{exp}} \in \{0.01, 0.05, 0.1, 0.2\}$; candidate budgets $M \in \{20, 100, 500\}$; interference radii $r \in \{0, 1, 2\}$; temporal persistence values

yielding short, medium, and long correlation; and sample sizes spanning at least one order of magnitude. Pareto/Lomax parameters must be chosen so that the asserted moment exists and the next higher diagnostic moment can fail. A bounded-outcome weak-overlap generator and a strong-overlap heavy-outcome generator are both required.

The planted hidden subgroup should depend on an interaction not represented by the fixed slices, for example

$$\mathbf{1}\{G = 1,\ \deg(v) > q_{0.9},\ \text{new-user} = 1,\ Z_t = \text{high-load}\}. \tag{45}$$

Its support and effect are varied independently. Discovery models are constrained by a minimum support rule; certification evaluates both the planted effect and false discoveries under a null generator.

J.2 Open Bandit protocol

Use one logged policy as behavior data and estimate the value of the other policy. The on-policy mean from the target log is computed on a disjoint time range when possible. Repeat in both directions and across policy mixtures. Report overlap diagnostics, effective sample size, and any context strata where the target action has inadequate support. Binary reward makes this a clean test of importance-weight tails but not of outcome-heavy-tail robustness.

J.3 TGB semi-synthetic protocol

Use the original timestamps and node identities. The logging score may be generated from a fixed temporal-graph backbone or a transparent structural score; the counterfactual estimator should not be credited for backbone accuracy. Potential outcomes depend on source/destination features, latent communities, direct exposure, and neighborhood exposure. Candidate-set construction is fixed before comparing estimators. Ground truth is obtained by replaying the known outcome function under target interventions on the same context stream.

J.4 KuaiRand protocol

Preserve user-level chronology and build folds by user/time blocks. Analyze binary interactions and watch time separately. Use the random-exposure records for intervention-oriented validation and the standard-feed records for weak-overlap stress tests. Claims must distinguish exact randomization-based checks from model-assisted extrapolation to target policies not represented by the random exposure mechanism.

J.5 Coverage computation

For each configuration and estimator, generate R independent repetitions. Empirical coverage is the fraction of intervals containing the ground truth; Wilson or Clopper–Pearson intervals quantify Monte Carlo uncertainty. A method with a high abstention rate is not assigned artificial coverage: coverage conditional on non-abstention and overall decision error are reported together. The final plots include a nominal-versus-empirical coverage curve and support-stratified coverage.

K Relation to COPF and Reuse of the Earlier Draft

The earlier project studied counterfactual online performative fairness for exposure in evolving-graph link recommendation. Its useful ingredients map into the present framework as follows.

Earlier component	Role in COMET
Shown/not-shown potential outcomes	One binary exposure mapping $S \in \{0, 1\}$
Graph-aware doubly robust pseudo-outcomes	Special case of Equation (3)
Residual-OI over group/score slices	Counterfactual moments with fixed auditors
Local-interference dependency graph	One source of the general conflict graph
Exposure-effect parity and calibration	Two examples in Table 1
Primal–dual decision controller	Baseline; replaced by certificate-guided repair
Strict propensity overlap	Replaced by target-support positivity and finite-p moments
Random-graph generative task	Optional protocol/design corollary, not a main contribution

The new paper should cite the earlier work in third person and explicitly state the difference in the introduction and appendix. Reusing notation or a special-case lemma is not enough to establish novelty; the main theorem, estimator, learned-auditor protocol, policy-repair result, and experiments must answer the general moment problem stated here.

L LIMITATIONS AND PRE-SUBMISSION CLAIM AUDIT

Before submission, every item below should be resolved in the manuscript or removed from the claimed contributions.

1. Prove the exact conditional version of Theorem 4.2 used by the implementation, including coupling constants and the adaptive scale rule.
2. Verify that the conflict graph corresponds to the synthetic and real-data preprocessing; report its construction algorithm, not only $\bar{\chi}_W$.
3. Show empirical coverage, not merely RMSE, across weak-overlap and outcome-tail regimes.
4. Demonstrate that graph/time corrections improve calibration of intervals rather than only widening them.
5. Validate the lower-bound assumptions and avoid presenting the cluster subclass as a lower bound for a narrower structured graph class without justification.
6. Report nuisance error in overlap-weighted norms or controlled misspecification sweeps.
7. Keep discovery and certification data separated through nuisance fitting, auditor selection, and hyperparameter tuning.
8. Replace every placeholder in Table 3; remove the table if the full comparison is not run.
9. Check all dataset licenses, outcome definitions, and intervention claims.
10. Recompile with the exact official ICLR 2027 style files, verify the nine-page main-text limit, anonymity, references, and the required AI use statement.